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Analysis of the AWS US-EAST-1 Outage of October 19 - 20, 2025

Introduction

Cloud computing has become the backbone of modern digital infrastructure, enabling applications to scale globally while relying on centralized service providers. However, this centralization also introduces risk, as failures within a major provider can cascade across industries. The Amazon Web Services (AWS) outage affecting the US-EAST-1 region on October 19 - 20, 2025, provides a strong example of how network dependencies, DNS failures, and service interconnections can lead to widespread disruption. Although many users experienced the outage on October 20, AWS reported that the incident began late on October 19 and continued into the following day. The root issue involved a failure in DNS resolution tied to Amazon DynamoDB, which triggered failures across multiple AWS services and downstream applications (Amazon Web Services [AWS], 2025).

Incident Summary

The outage began at approximately 11:48 PM PDT on October 19, 2025, when AWS detected increased API error rates in the US-EAST-1 region. By 12:26 AM PDT on October 20, AWS identified the issue as related to DNS resolution failures affecting DynamoDB endpoints. Specifically, a race condition in AWS's DNS management system resulted in an empty DNS record for the regional endpoint, preventing clients from resolving the service address (AWS, 2025).

AWS restored DNS records by 2:25 AM PDT, and services gradually recovered as DNS caches updated. However, the incident continued to affect other services throughout the day. Issues with EC2 instance launches, network propagation delays, and Network Load Balancer (NLB) health checks persisted. AWS reported significant recovery by 12:28 PM PDT, with full restoration completed by 3:01 PM PDT (AWS, 2025).

The outage impacted numerous downstream applications, including Snapchat, Roblox, Fortnite, Venmo, Zoom, Slack, Pinterest, and Duolingo (The Verge, 2025). These services depend on AWS for backend infrastructure such as databases, APIs, authentication systems, and content delivery. For example, a platform like Venmo relies on AWS for secure transaction processing and API communication, while gaming services such as Roblox depend on cloud-hosted servers to manage user sessions and gameplay. When AWS services fail, these applications cannot function properly, even if their internal code remains operational.

Technical and Network Analysis

This incident highlights the importance of several networking concepts, particularly DNS resolution, application-layer protocols, and transport-layer communication. DNS operates at the application layer and is responsible for translating domain names into IP addresses. In this outage, the failure of DNS resolution meant that applications could not locate the DynamoDB service endpoint, effectively blocking communication before it could begin (AWS, 2025).

Once DNS resolution failed, higher-level communication protocols such as HTTPS over TCP (port 443) were also impacted. Even though the network infrastructure itself remained operational, applications experienced latency, connection failures, and increased error rates because they could not establish connections to required services. This demonstrates how failures at the application layer can disrupt the entire communication stack.

The outage also revealed the interconnected nature of cloud services. DynamoDB serves as a foundational component for many AWS services, including EC2. AWS reported that EC2's internal systems rely on DynamoDB for state management and lease tracking. When DynamoDB became unavailable, EC2 instance launches failed, and network configuration processes were delayed. Additionally, Network Load Balancers began misinterpreting instance health, leading to further connection errors (AWS, 2025).

From a networking perspective, this incident illustrates that failures in one layer can propagate across multiple layers and services. DNS, TCP/IP communication, load balancing, and service orchestration are tightly coupled in cloud environments. A failure in service discovery can therefore trigger cascading failures throughout the system.

Service and Business Impact

The AWS outage affected a wide range of industries, including social media, gaming, financial services, communication platforms, and e-commerce. According to Reuters (2025), the disruption impacted everyday activities such as payments, messaging, and online transactions. This demonstrates the extent to which modern society depends on cloud infrastructure.

Reliance on a single cloud provider, particularly within a dominant region like US-EAST-1, creates systemic risk. Many organizations assume high availability due to AWS's scale, but they may still depend on a single region for critical services such as databases or authentication systems. When that region experiences an outage, multiple dependent services fail simultaneously.

Although exact financial losses are not publicly available for all affected companies, reasonable estimates can be made. For example, a company generating $500,000 per day would lose

approximately $20,833 per hour. A six-hour outage could therefore result in over $125,000 in lost revenue, excluding additional costs such as customer dissatisfaction, refunds, and reputational damage.

In an academic context, similar risks apply to platforms such as D2L Brightspace. If a cloud-dependent learning management system experiences latency or downtime during an exam, students may be unable to submit assignments or access test materials. This could lead to grading inconsistencies, delays, and fairness concerns. Institutions must therefore consider cloud reliability when designing digital learning environments.

Lessons Learned and Recommendations

The AWS outage demonstrates several key lessons about network design and cloud reliability. First, DNS is a critical dependency in modern systems. Without proper DNS resolution, applications cannot function even if all other infrastructure components are operational. Second, the incident highlights the interdependence of cloud services, where failures in one component can propagate across multiple systems. Third, it emphasizes that large-scale cloud providers are not immune to failure, and organizations must design systems with resilience in mind.

To reduce similar risks, organizations should implement the following recommendations:

1. Adopt Multi-Region Architectures

2. Critical applications should be deployed across multiple regions to ensure availability during regional outages.

3. Identify and Eliminate Single Points of Failure

Organizations should analyze their infrastructure dependencies, particularly around DNS, databases, and authentication services, to ensure redundancy.

4. Implement Graceful Degradation Strategies

Systems should be designed to maintain partial functionality during outages. For example, an LMS could allow offline submissions or automatically extend deadlines during disruptions.

These strategies can significantly reduce the impact of infrastructure failures and improve overall system resilience.

Conclusion

The AWS US-EAST-1 outage of October 19 - 20, 2025, demonstrates how a seemingly small issue, such as a DNS resolution failure, can escalate into a global disruption affecting multiple industries. The incident highlights the importance of DNS, network protocols, and service dependencies in cloud computing environments. It also underscores the need for organizations to design systems that anticipate failure rather than assume constant availability. By implementing resilient architectures and proactive monitoring strategies, organizations can reduce the risks associated with cloud dependence and ensure more reliable service delivery.

References

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